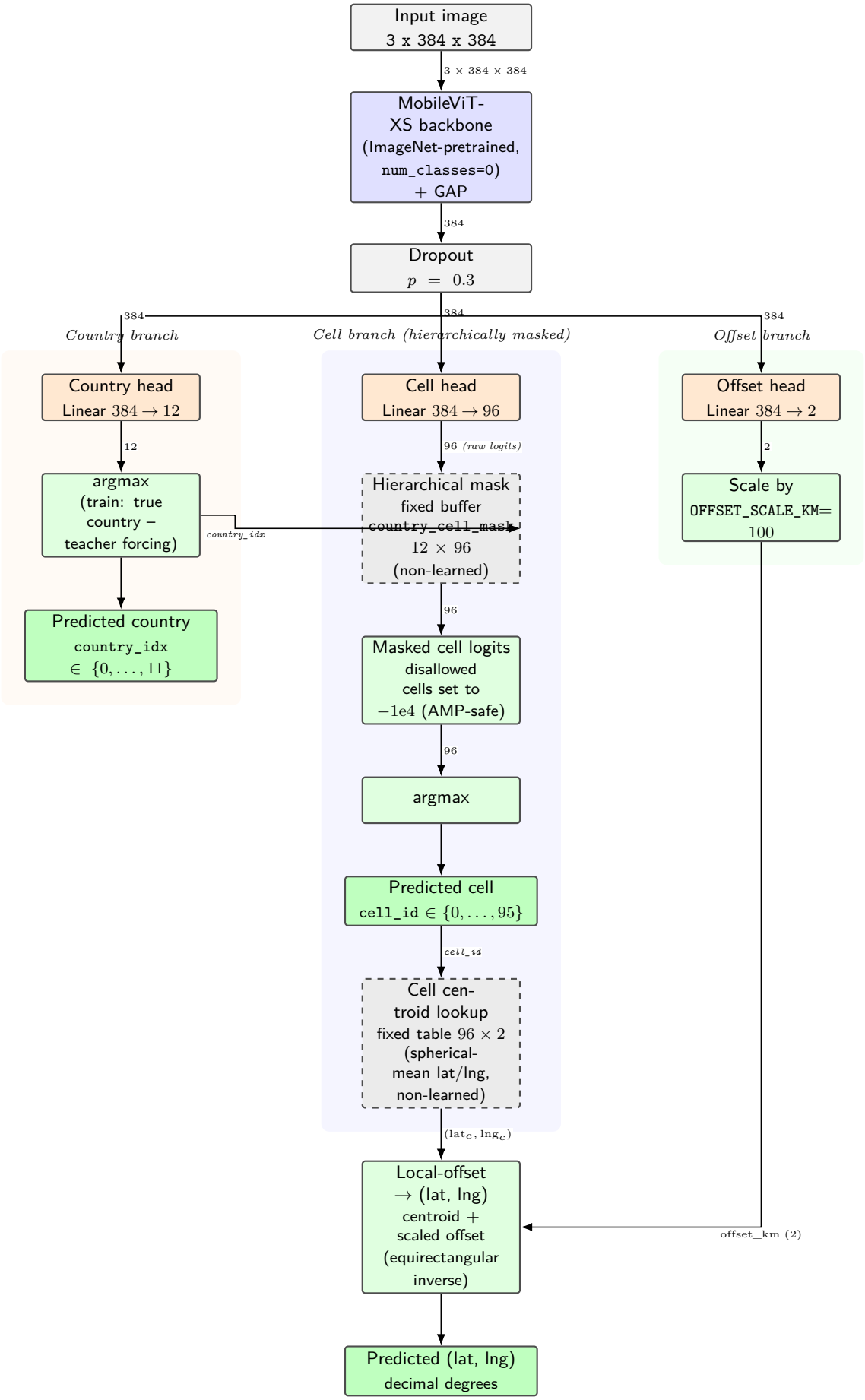
Input image
3 x 384 x 384
3 × 384 × 384
MobileViT-XS backbone (ImageNet-pretrained, num_classes=0) + GAP
384
Dropout
p = 0.3
384
384
384
Country branch
Cell branch (hierarchically masked)
Offset branch
Country head
Linear 384 → 12
12
argmax
(train: true country – teacher forcing)
country_idx
Predicted country
country_idx ∈ {0, . . . , 11}
Cell head
Linear 384 → 96
96 (raw logits)
Hierarchical mask
fixed buffer country_cell_mask
12 × 96
(non-learned)
96
Masked cell logits
disallowed cells set to −1e4 (AMP-safe)
96
argmax
Predicted cell
cell_id ∈ {0, . . . , 95}
cell_id
Cell centroid lookup
fixed table 96 × 2
(spherical-mean lat/lng, non-learned)
(lat_c, lng_c)
Offset head
Linear 384 → 2
2
Scale by
OFFSET_SCALE_KM= 100
offset_km (2)
Local-offset → (lat, lng)
centroid + scaled offset
(equirectangular inverse)
Predicted (lat, lng)
decimal degrees